Dominique Piché-Meunier

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EXPERIENCE

Machine Learning Engineer – GenAI and Computer Vision 2024–now
Adobe, Substance 3D Tech Transfer Team, San Francisco, CA

- Trained and deployed large-scale diffusion models for 3D-aware image generation, conditioning on depth, lighting, and camera parameters to enable seamless integration of 3D content into 2D imagery
- Led the data backend for Adobe's 3D GenAI research, building a Kubernetes-based compute platform on AWS scaling to tens of thousands of concurrent jobs, supporting the training of 3D technology deployed in Illustrator, Substance API, Photoshop
- Designed and maintained automated pipelines to process large-scale 3D datasets (meshes, Gaussian splats, materials) with PostgreSQL and AWS services, supporting over 100M images and videos

Research Intern – Machine Learning and Computer Vision 2022–2023
Adobe, Substance 3D Tech Transfer Team, San Francisco, CA

- Owned the full ML lifecycle : synthetic data generation, model development, fine-tuning, evaluation, and deployment into Adobe product environments
- Developed novel computer vision models for 3D content manipulation (geometry reconstruction, lighting estimation), leading to an ICCV publication and a patent
- Collaborated with product managers and designers to refine models based on user feedback, optimizing models for on-device inference and product integration

Research Intern – Machine Learning and Computer Vision 2020–2021
Institute Intelligence and Data (IID), Université Laval, Québec

- Generated a large-scale synthetic dataset by compositing 3D objects into real scenes with Blender, supporting robust training of geometry and camera estimation models.
- Trained neural networks to estimate camera intrinsics and extrinsics from a single image, and conducted user studies to evaluate calibration accuracy from a perceptual standpoint
- Led the preparation and writing of a TPAMI submission

EDUCATION

M.S. in Electrical Engineering 2021-2023
Université Laval, Québec
Thesis : Deep single image depth of field estimation
Supervisors : Jean-François Lalonde (Université Laval) and Yannick Hold-Geoffroy (Adobe Research)
GPA : 4.33/4.33

B.S. in Engineering Physics 2017-2021
Université Laval, Québec
GPA : 4.21/4.33

PUBLICATIONS

Dominique Piché-Meunier, Yannick Hold-Geoffroy, Jianming Zhang, Jean-François Lalonde. *Lens Parameter Estimation for Realistic Depth of Field Synthesis*. International Conference on Computer Vision (ICCV), 2023.

François-Alexandre Tremblay, Dominique Piché-Meunier, Louis Dubois. *Multi-objective optimization for designing salary structures*. International Conference on the Integration of Constraint Programming, Artificial Intelligence, and Operations Research (CPAIOR), 2022.

Yannick Hold-Geoffroy, Dominique Piché-Meunier, Kalyan Sunkavalli, Jean-Charles Bazin, François Rameau, Jean-François Lalonde. *A Deep Perceptual Measure for Lens and Camera Calibration*. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2021.

Erik Buch Jørgensen, Jacob Graversen Johansen, Joakim Overgaard, Dominique Piché-Meunier, Daline Tho, Haydee Maria Linares Rosales, Kari Tanderup, Luc Beaulieu, Gustavo Kertzscher, Sam Beddar. *A high-Z inorganic scintillator–based detector for time-resolved in vivo dosimetry during brachytherapy*. Medical Physics, 2021.

Jacob Graversen Johansen, Erik Buch Jørgensen, Joakim Overgaard, Dominique Piché-Meunier, Haydee Maria Linares Rosales, Daline Tho, Kari Tanderup, Sam Beddar, Luc Beaulieu, Gustavo Kertzsche. *Characterisation of an inorganic scintillation detector system for time resolved in vivo dosimetry*. World Congress of Brachytherapy, 2021.

SCHOLARSHIPS

- **NSERC Canada Graduate Scholarships - Master's** 2021
  For academic excellence and research potential *17500$*
- **FRQNT Québec Graduate Scholarships - Master's** 2021
  For academic excellence and research potential *35000$*
- **NSERC Undergraduate Student Research Awards** 2020
  For academic excellence *4500$*
- **FRQNT Supplements of the Undergraduate Student Research Awards** 2020
  For academic excellence *1500$*
- **NSERC Undergraduate Student Research Awards** 2019
  For academic excellence *4500$*
- **FRQNT Supplements of the Undergraduate Student Research Awards** 2019
  For academic excellence *2000$*
- **Hydro-Québec Entrance Scholarship** 2017
  For academic excellence *3000$*
- **Franco Rasetti Entrance Scholarship** 2017
  For academic excellence *1500$*

LANGUAGES

Languages : french (native), english (fluent)